

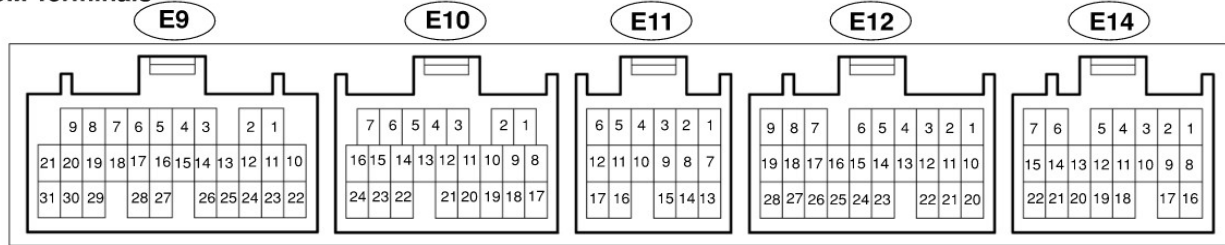


Year  
**4Runner:** 1999-2000  
**Tacoma:** 2000-2001  
**Engine:** 5VZ-FE (3.4L V6)  
**Board:** uaEFI Adapter Board V0.2

**Notes:**  
 California or Federal spec emissions (one or two catalytic converters from factory)  
 2000: California-style emissions (two catalytic converters from factory and upstream Air Fuel Ratio sensor)

**\*\*WARNING\*\*** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

## ECM Terminals



Plug "A"

Plug "B"

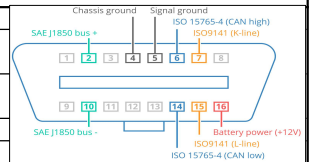
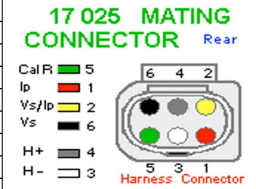
Plug "C"

Plug "D"

Plug "E"

### OPEN PINS

| Note: Populate open slots in ECU connectors with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 82998-12740) pins                                                                                                                                                                                                                                  |      |     |                           |                           |                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|---------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------|
| Connector                                                                                                                                                                                                                                                                                                                                                           | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                         |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 5   | Lowside Output (with PWM) | B17 Low Side Output 3     | PWM Ground Output                                                                                             |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 17  | Ignition Output 4         | B11 Coil 4                | **MUST be used with an igniter or smart coils**                                                               |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 18  | Ignition Output 5         | B13 Coil 5                | **MUST be used with an igniter or smart coils**                                                               |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 19  | Ignition Output 6         | B10 Coil 6                | **MUST be used with an igniter or smart coils**                                                               |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 23  | G-                        | N/A                       | For separated negative VR leads in the factory wiring harness. Connect camshaft VR- here.                     |
| E9                                                                                                                                                                                                                                                                                                                                                                  | A    | 24  | +B                        | N/A                       | +12V battery voltage                                                                                          |
| Connector                                                                                                                                                                                                                                                                                                                                                           | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                         |
| E10                                                                                                                                                                                                                                                                                                                                                                 | B    | 1   | HB1+                      |                           | H-Bridge 1 +                                                                                                  |
| E10                                                                                                                                                                                                                                                                                                                                                                 | B    | 3   | HB1-                      |                           | H-Bridge 1 -                                                                                                  |
| E10                                                                                                                                                                                                                                                                                                                                                                 | B    | 11  | PPS2                      | C4 PPS2                   | Pedal Position Sensor 2 (Can be used for any 0-5v Analog Input)                                               |
| E10                                                                                                                                                                                                                                                                                                                                                                 | B    | 13  | TPS2                      | C14 TPS2                  | Throttle Position Sensor 2 (Can be used for any 0-5v Analog Input)                                            |
| E10                                                                                                                                                                                                                                                                                                                                                                 | B    | 20  | PPS1                      | D6 PPS1                   | Pedal Position Sensor 1 (Can be used for any 0-5v Analog Input)                                               |
| Connector                                                                                                                                                                                                                                                                                                                                                           | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                         |
| <b>**WARNING**</b> If your vehicle has a factory 17-pin center connector, do <b>NOT</b> plug it into this adapter board! Leave it disconnected! It is for the immobilizer system and is no longer used<br>Use a MLC040IIHD-17S (OEM p/n 353027-1) 17-pin connector and populate with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 316838-1) pins |      |     |                           |                           |                                                                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 1   | IP                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 1)                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 2   | H-                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO Heater - (Connector Pin 3)                                                      |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 3   | CalR                      | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 5)                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 4   | 7th Injector              | B9 MR Weak LS 1           | Connect 7th injector negative here, and positive to +B (12V battery)                                          |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 5   | MAP                       | D9 MAP                    | Manifold Absolute Pressure Sensor (Can be used for any 0-5v Analog Input)                                     |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 6   | AUX2                      | C3 AUX2                   | Analog Input #2 (0-5v)                                                                                        |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 7   | VS/IP                     | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 2)                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 8   | H+                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO Heater + (Connector Pin 4)                                                      |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 9   | VS                        | N/A                       | For Bosch 17025 (LSU 4.9) WBO (Connector Pin 6)                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 10  | 5V Sensor                 | N/A                       | +5V for sensors                                                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 11  | 5V Sensor                 | N/A                       | +5V for sensors                                                                                               |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 12  | Flex                      | D5 FLEX                   | Flex Fuel Signal                                                                                              |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 13  | +B                        | N/A                       | +12V battery voltage                                                                                          |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 14  | +B                        | N/A                       | +12V battery voltage                                                                                          |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 15  | AUX1                      | D1 AUX1                   | Analog Input #1 (0-5v)                                                                                        |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 16  | Sensor Ground             | N/A                       | For sensor grounds only                                                                                       |
| E11                                                                                                                                                                                                                                                                                                                                                                 | C    | 17  | Sensor Ground             | N/A                       | For sensor grounds only                                                                                       |
| Connector                                                                                                                                                                                                                                                                                                                                                           | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                         |
| E12                                                                                                                                                                                                                                                                                                                                                                 | D    | 1   | Digital Input 2           | MEGA_EPIC_D30             | Ground activated                                                                                              |
| E12                                                                                                                                                                                                                                                                                                                                                                 | D    | 5   | Digital Input 1           | MEGA_EPIC_D31             | Ground activated                                                                                              |
| Connector                                                                                                                                                                                                                                                                                                                                                           | Plug | Pin | Function                  | Tuner Studio Channel Name | Notes                                                                                                         |
| E14                                                                                                                                                                                                                                                                                                                                                                 | E    | 18  | CAN1H                     | N/A                       | CANBUS High. If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #6 |
| E14                                                                                                                                                                                                                                                                                                                                                                 | E    | 19  | CAN1L                     | N/A                       | CANBUS Low. If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #14 |
| E14                                                                                                                                                                                                                                                                                                                                                                 | E    | 20  | Lowside 1                 | MEGA_EPIC_D45             | On/Off Ground Output                                                                                          |
| E14                                                                                                                                                                                                                                                                                                                                                                 | E    | 21  | Lowside 2                 | MEGA_EPIC_D46             | On/Off Ground Output                                                                                          |



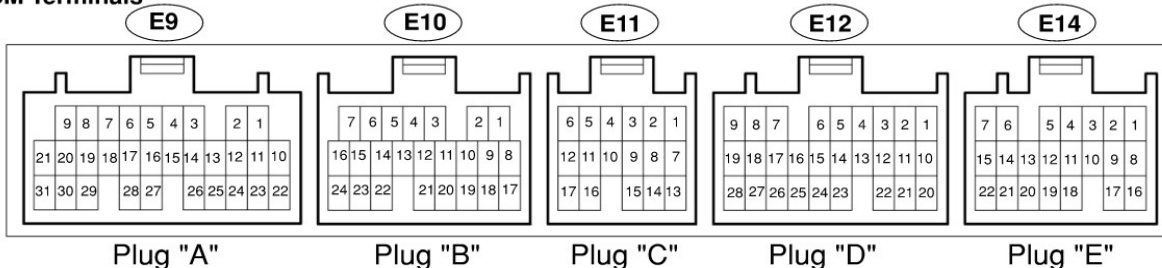


**Year**  
**4Runner:** 1999-2000  
**Tacoma:** 2000-2001  
**Engine:** 5VZ-FE (3.4L V6)  
**Board:** uaEFI Adapter Board V0.2

**Notes:**  
 California or Federal spec emissions (one or two catalytic converters from factory)  
 2000: California-style emissions (two catalytic converters from factory and upstream Air Fuel Ratio sensor)

**\*\*WARNING\*\*** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

## ECM Terminals



## POPULATED PINS

**Note:** Populate open slots in ECU connectors with MLC040III-SKT1 (OEM p/n 82998-12690) or MLC040III-SKT2 (OEM p/n 82998-12740) pins

| Connector | Plug | Pin | Function | Tuner Studio Channel Name | Notes                                                                             |
|-----------|------|-----|----------|---------------------------|-----------------------------------------------------------------------------------|
| E9        | A    | 1   | #30      | B4 Injector Output 3      | Injector #3                                                                       |
| E9        | A    | 2   | #40      | B3 Injector Output 4      | Injector #4                                                                       |
| E9        | A    | 3   | #50      | B2 Injector Output 2      | Injector #5                                                                       |
| E9        | A    | 4   | #60      | B1 Injector Output 6      | Injector #6                                                                       |
| E9        | A    | 7   | S1       | MEGA_EPIC_D39             | Shift Solenoid 1                                                                  |
| E9        | A    | 8   | S2       | MEGA_EPIC_D41             | Shift Solenoid 2                                                                  |
| E9        | A    | 9   | SL       | MEGA_EPIC_D5              | TCC Lockup Solenoid                                                               |
| E9        | A    | 10  | G2       | C7 HALL3                  | Cam Signal                                                                        |
| E9        | A    | 11  | IGT1     | B15 Coil 1                | Ignition Coil #1                                                                  |
| E9        | A    | 12  | IGT2     | B14 Coil 2                | Ignition Coil #2                                                                  |
| E9        | A    | 13  | IGT3     | B12 Coil 3                | Ignition Coil #3                                                                  |
| E9        | A    | 14  | SP2+     | C18 VR1+ Discrete         | Speed Sensor 2                                                                    |
| E9        | A    | 15  | RSC      | B18 Low Side Output 2     | Idle Air Control Valve - Close                                                    |
| E9        | A    | 16  | RSO      | B7 Low Side Output 1      | Idle Air Control Valve - Open                                                     |
| E9        | A    | 22  | PWR      | MEGA_EPIC_D24             | ECT PWR Button (Automatic Transmission)                                           |
| E9        | A    | 27  | KNK1     | D14 Knock Input           | Knock Sensor Bank 1 (use selector switch on carrier board to choose KNK1 or KNK2) |
| E9        | A    | 28  | KNK2     | D14 Knock Input           | Knock Sensor Bank 2 (use selector switch on carrier board to choose KNK1 or KNK2) |
| E10       | B    | 4   | HTAF     | MEGA_EPIC_D13             | Upstream O2 Sensor Heater                                                         |
| E10       | B    | 5   | #10      | B6 Injector Output 1      | Injector #1                                                                       |
| E10       | B    | 6   | #20      | B5 Injector Output 2      | Injector #2                                                                       |
| E10       | B    | 7   | EVP1     | MEGA_EPIC_D3              | EVAP Solenoid                                                                     |
| E10       | B    | 9   | PSW      | MEGA_EPIC_D27             | Power Steering Switch                                                             |
| E10       | B    | 10  | VG       | C15 AUX3                  | Mass Air Flow Sensor Signal                                                       |
| E10       | B    | 12  | OX1+     | MEGA_EPIC_ADC_0           | Upstream Air Fuel Ratio Sensor Signal                                             |
| E10       | B    | 12  | OX1      | MEGA_EPIC_ADC_1           | Upstream Narrowband Oxygen Sensor Signal                                          |
| E10       | B    | 14  | THW      | D16 CLT Coolant           | Engine Coolant Temperature                                                        |
| E10       | B    | 15  | OIL      | MEGA_EPIC_ADC_12          | Automatic Transmission Fluid Temperature                                          |
| E10       | B    | 16  | NE+      | C6 HALL2                  | Crank Sensor                                                                      |
| E10       | B    | 22  | THA      | D15 IAT                   | Intake Air Temperature                                                            |
| E10       | B    | 23  | VTA      | D13 TPS1                  | Throttle Position Sensor                                                          |
| E12       | D    | 2   | R        | MEGA_EPIC_D34             | Reverse (Automatic Transmission)                                                  |
| E12       | D    | 3   | 2        | MEGA_EPIC_D35             | 2nd Gear Selector (Automatic Transmission)                                        |
| E12       | D    | 4   | IDLO     | MEGA_EPIC_D40             | Idle Sensor                                                                       |
| E12       | D    | 6   | ODMS     | MEGA_EPIC_D37             | Overdrive Lockout Switch / Light                                                  |
| E12       | D    | 7   | OILW     | MEGA_EPIC_D42             | A/T Fluid Overheat Light                                                          |
| E12       | D    | 8   | OXS      | MEGA_EPIC_ADC_15          | Downstream Oxygen Sensor                                                          |
| E12       | D    | 9   | HTS      | MEGA_EPIC_D12             | Downstream Oxygen Sensor Heater                                                   |
| E12       | D    | 10  | OD2/ODLP | MEGA_EPIC_D37             | Overdrive Lockout Switch / Light                                                  |
| E12       | D    | 11  | PWR      | MEGA_EPIC_D24             | ECT PWR Button (Automatic Transmission)                                           |
| E12       | D    | 12  | L        | MEGA_EPIC_D33             | Low Gear Selector (Automatic Transmission)                                        |
| E12       | D    | 13  | ACT      | MEGA_EPIC_D49             | A/C Terminate                                                                     |
| E12       | D    | 18  | 4WD/AWD  | MEGA_EPIC_D29             | 4WD Indicator                                                                     |
| E12       | D    | 19  | L4       | MEGA_EPIC_D25             | Transfer Case Low Range                                                           |
| E12       | D    | 20  | NSW      | MEGA_EPIC_D32             | Neutral Safety Switch                                                             |
| E12       | D    | 22  | SP1      | C5 HALL1                  | Speed Sensor 1                                                                    |
| E12       | D    | 25  | AC1      | MEGA_EPIC_D23             | A/C Request                                                                       |
| E14       | E    | 3   | FC       | B16 Low Side Output 4     | Fuel Pump                                                                         |
| E14       | E    | 6   | W        | MEGA_EPIC_D43             | Check Engine Light                                                                |
| E14       | E    | 7   | STA      | MEGA_EPIC_D28             | Cranking Signal                                                                   |
| E14       | E    | 15  | STP      | MEGA_EPIC_D22             | Brake Light                                                                       |
| E14       | E    | 17  | PTNK     | MEGA_EPIC_ADC_14          | EVAP Pressure                                                                     |